A COMPUTATIONAL DESIGN SCIENCE APPROACH FOR INTER-VEHICLE NETWORK COMMUNICATION USING LEO SATELLITE INTERNET

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Abstract

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Declarations

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1 Introduction

1.1 CBRN Defense Field

CBRN defense field encompasses the military and civil measures taken to protect against chemical, biological, radiological, and nuclear threats. The field includes a wide range of activities, such as detection, identification, decontamination, and medical countermeasures (Otřísal, 2014). CBRN events can pose a significant physical and psychological hazard to the public and pose environmental and economic impact (Ministry of Health Malaysia, 2025). Fast, coordinated and accurate response in the first critical hour or two is crucial to minimize the impact of such events (Joint Emergency Services Interoperability Programme (JESIP), 2016). Providing accurate information in a timely manner is essential for effective decision-making and response (Ministry of Health Malaysia, 2025).

Coordination between the various agencies and personnel involved in CBRN defense is essential. Numerous software solutions have been developed to facilitate communication and information sharing among CBRN defense teams (Marques, Duarte & Lobo, 2025; Environics Oy, 2026; Bruhn NewTech, 2021; Observis Oy, 2026). A common pattern among these solutions is the near real-time consolidation and visualization of data from multiple sources, such as sensors, first responders, command centers and various governmental databases. However, the effectiveness of these solutions is limited by the communication infrastructure available in the affected area. Additionally, any delays introduced by the communication infrastructure, protocols, and data processing can add up to significant delays in the overall response time. In the case of CBRN events, where every second counts, these delays can have serious consequences.

1.2 LEO Satellite Networks

One of the more recent developments in communication infrastructure is the commercial availability of low earth orbit (LEO) satellite internet. LEO satellite networks have the potential to provide high-speed, low-latency internet access to remote and underserved areas (Duan & Dinavahi, 2021). This technology has the potential to improve the communication abilities of CBRN defense teams by providing a communication channel in remote areas or areas with delibierately disrupted communication infrastructure (Abels, 2024).

One of the leading providers of LEO satellite internet is Starlink, which as of September 2024 has launched 6426 satellites into orbit and has 14 million active subscribers globally, making it the leader by a wide margin in both of these metrics. Starlink subscriptions are widely available across the globe in both personal and business plans (Koosha, Madani &

Ardakani, 2025). The relatively low subscription cost, easy of acquisition, low-latency, high bandwidth and coverage make it a compelling option as the communication infrastructure for CBRN defense teams.

1.3 QUIC Protocol

QUIC is a multiplexed, encrypted and low-latency transport protocol that is designed with the goal of improving transport performance and security of internet applications. QUIC is a cross-layer protocol that operates at the Transport, Security and Application layers of the OSI model (Langley et al., 2017).

According to the designers of the QUIC protocol, two of the main motivators behind its design were reducing the handshake delay of TCP and TLS and Head-of-line blocking delay within TCP. Additionally, TCP is a rigid protocol which cannot be modified to suit the needs of modern applications or specific use cases. QUIC comes out as an answer to these limitations and by measurements from Google it reduces latency of Google searches by 8% on desktop and by 3.6% on mobile. (Langley et al., 2017) The flexibility and low latency characteristics of QUIC makes it particularly suitable for the demanding application of CBRN monitoring and communication over LEO satellite networks.

1.4 Motivation and Context

This thesis aims to address the challenges of near real-time communication for CBRN applications with global coverage by proposing a computational design science approach for inter-vehicle network communication using QUIC as a transport protocol over LEO satellite internet. The proposed approach leverages the advantages of LEO satellite internet, such as low latency and high bandwidth, to enhance the communication capabilities of CBRN defense vehicles. QUIC is used as a transport protocol due to its flexibility for specific applications and low latency. By improving the speed and accuracy of information sharing, the proposed approach has the potential to significantly enhance the effectiveness of CBRN defense operations. The proposed design aims to enable wider capabilities of software systems for CBRN defense, such as global connectivity in remote areas and reduced latency and reliability over traditional network infrastructure (e.g. 3G, 4G, LTE, 5G and geostationary satellite internet) and transport protocols.

1.5 Objectives and Scope of This Work

This thesis adopts the Design Science Research Methodology (DSRM) framework by Peffers et al. (2007) to design, implement, and evaluate a proof-of-concept (PoC) software communication artifact. The primary aim of the artifact is to leverage the QUIC transport protocol

over Low Earth Orbit (LEO) satellite internet for near real-time CBRN communication. Furthermore, Machine Learning (ML) will be used to optimize the performance of the QUIC protocol over LEO satellite networks, with a focus on minimizing latency, improving resilience and prioritizing critical CBRN data payloads. The artifact will be evaluated in a real-world environment using Starlink's commercial LEO satellite network.

To achieve this aim, the work is divided into the following concrete research objectives:

- **O1 (Artifact Design and Implementation):** Develop a software prototype capable of transmitting and receiving CBRN data payloads between remote vehicular nodes using the QUIC transport protocol with ML optimizations.
- **O2 (Payload Simulation):** Design a standardized, synthetic data schema that mimics real-world CBRN sensor telemetry (e.g., detection alerts, GPS coordinates, and timestamped hazard measurements) without relying on proprietary hardware.
- **O3 (Empirical Evaluation):** Measure the performance of the implemented QUIC artifact over a live commercial LEO satellite network (Starlink) under real-world operating conditions, tracking metrics such as end-to-end latency, handshake duration, packet loss, and connection recovery during satellite handovers.
- **O4 (Comparative Benchmarking):** Compare the performance of the QUIC-based artifact with and without ML optimizations against a traditional baseline protocol stack (i.e., TCP with TLS 1.3) running over the same LEO satellite network.

To guide the empirical evaluation of the artifact, this study formulates two core research questions (RQs) and their corresponding testable hypotheses (H_1 and H_2):

> **RQ1:** How does replacing traditional TCP/TLS transport protocols with standard QUIC affect the latency, connection resilience, and real-time data delivery of CBRN telemetry when routed over commercial LEO satellite networks?
>
> **Hypothesis 1 (H_1):** Because of its 0-RTT/1-RTT handshake mechanism and elimination of head-of-line blocking, utilizing the standard QUIC protocol over LEO satellite internet significantly reduces end-to-end transmission latency and improves connection recovery times during network dropouts compared to a standard TCP/TLS baseline.
>
> **RQ2:** To what extent does applying Machine Learning (ML) optimizations to the QUIC protocol further enhance packet prioritization and resilience during predictable LEO network degradations, such as satellite handovers?

Hypothesis 2 (H_2): An ML-optimized QUIC implementation will yield a statistically significant reduction in latency for high-priority CBRN payloads during satellite handovers compared to a non-optimized QUIC baseline, by dynamically adapting transport parameters to the fluctuating LEO network state.

Delimitations and Scope: To maintain scientific rigor and feasibility within the timeframe of a Master's thesis, explicit boundaries are set regarding the scope of this study:

- **Network Infrastructure:** The empirical testing is strictly limited to the Starlink commercial LEO satellite constellation. Other LEO constellations (such as OneWeb or Amazon Kuiper), geostationary (GEO) satellites, and terrestrial cellular networks (e.g. 3G, 4G, LTE, 5G) are excluded from the experimental setup.
- **Hardware and Sensors:** Physical CBRN detection hardware, chemical analyzers, and vehicular sensor integration are out of scope. The study relies entirely on software-simulated data payloads that accurately replicate the data volume and frequency of standard CBRN defense devices.
- **C4I Integration and UI:** While the artifact demonstrates data transport capabilities, developing a full Command, Control, Communications, Computers, and Intelligence (C4I) situational awareness dashboard or graphical user interface (GUI) is out of scope. The focus remains strictly on the networking and transport layers.
- **Adversarial Security Testing:** Although QUIC inherently enforces TLS 1.3 encryption, active electronic warfare testing—such as intentional radio frequency (RF) jamming, satellite spoofing, or deep cryptographic penetration testing—is excluded from this work.

2 Theoretical Background

The following sections provide a theoretical background of the various technologies, protocols, standards and scientific approaches relevant to this thesis work.

2.1 CBRN Defense Systems

CBRN defense systems are systems which are designed to aid first responders with their strategic decision making, standard operating procedures and situational awareness before, during and after the occurance of CBRN incidents. CBRN defense refers to the defense against chemical, biological, radiological and nuclear threats.

CBRN defense systems are used by militaries, first responders and dedicated CBRN defense units. These systems are comissioned by governments in accordance with CBRN preparedness goals and strategies. Such systems typically consist of hardware and software components which work together to collect, process and present information related to CBRN incidents. The hardware can include CBRN detectors, sensors, communication devices, personal computers, laboratory equipment and various mechanical and electrical automation systems. The exact hardware depends on the purpose of the system and the environment it is designed to operate in. (Policastro, 2008; Thompson & Gahlaut, 2015)

The software components are used to collect and proccess information from the hardware components and provide it to the users of the system. A common function of these types of systems is to provide near real-time measurements and alerts from collected sensors and other information sources. Furthermore, the information is typically presented to users using a map-based interface, where measurements, detected hazards, personnel, equipment and other relevant information are shown in reference to their geographical location. This allows information from multiple sources to be presented to users in a way that improves their situational awareness.

CBRN defense systems typically record all available information within the system by design, which enables users to performpost event analysis of CBRN incidents. Reviewing and cross referencing plots of recorded measurements and locations, generating and managing reports, overall incident review and incident documentation are common activities. These features can be typically designed in accordance with organizational standard operating procedures and regulations, reducing the need for manual documentation and reporting and increasing the accuracy of results from such procedures. Furthermore, the ability to perform post analysis of CBRN incidents can be used to improve organizational performance and preparedness for future incidents.

These systems need to provide information with low delay and sufficient accuracy for decision making. Reliability, promptness and availability are important features of these systems, as they may operate during situations in which the communication infrastructure or individual sub-components are unavailable or degraded. Each sub-component of the software system can introduce delays and inaccuracies, act as a bottleneck or a point of failure. Therefore, every layer of the system and interfaces between sub-components need to be designed with rigorous performance and reliability requirements.

2.2 Low Earth Orbit (LEO) Satellite Networks

Low Earth Orbit (LEO) satellites are satellites that orbit the Earth at a speed higher than that of Earth's rotation. As a result of that higher speed, LEO satellites appear to continuously circulate around the Earth. In order to provide global coverage a constellation of LEO satellites is required. (Sun, 2014)

The altitude of a satellite has a direct effect on the characteristics of the communication system. Compared to geostationary Earth orbit (GEO) satellites (see Figure 1), LEO satellites operate considerably closer to the Earth (see Figure 2). The shorter distance reduces the signal propagation time and the free-space path loss. This makes LEO satellites suitable for services where delay is important. The lower altitude also means that an individual satellite covers a smaller area and remains visible to a user for a limited time. Furthermore, a LEO network must use multiple satellites and coordinate the changing connectivity between satellites, user terminals and gateway stations. The resulting network is categorized as a mobile communication system rather than a single fixed satellite link of fixed infrastructure communication system. (Sun, 2014)

The following discussion of LEO satellite orbit mechanics, link characteristics and networking concepts draws primarily on the satellite networking principles presented by Sun (2014).

2.2.1 Orbital and Constellation Characteristics

An orbit is determined by the motion of a satellite under the influence of the Earth's gravity. The orbital altitude, inclination, eccentricity and position of the orbital plane determine the satellite's ground track and the areas that can be served. For communication systems, these parameters are selected together with the number and arrangement of satellites to provide the required coverage and availability. A single LEO satellite cannot provide continuous service over a large geographical area because its footprint moves across the Earths' surface. A constellation addresses this limitation by distributing satellites across several orbital planes so that another satellite can become available as the current satellite moves out of view. (Sun, 2014)

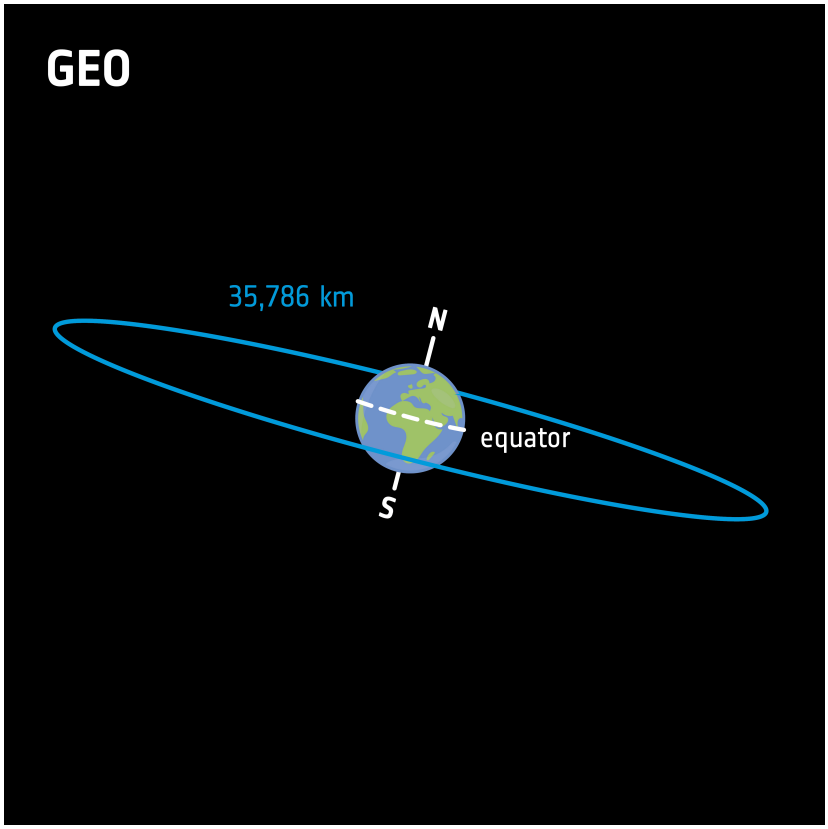


Figure 1: Illustration of geostationary orbit (GEO). Satellites in GEO orbit at approximately 35 786 km above Earth's equator, matching Earth's rotation to appear stationary over a fixed position on the ground. Image credit: ESA, reproduced under ESA standard licence for educational and informational use.

The movement of a LEO constellation is predictable, but the communication topology is dynamic. The relationships between satellites, gateways and user terminals change continuously as the satellites follow their orbits. In a constellation with inter-satellite links, the links within an orbital plane can remain relatively stable for a period of time, while links between orbital planes and links to ground stations change as the satellites move. This predictability can be used for network planning and routing, but it does not make the network static. Routing decisions and resource allocation must account for the expected lifetime and quality of available links. (Sun, 2014)

2.2.2 LEO Satellite Link Characteristics

The communication path between an application and a remote endpoint may contain several segments: the user terminal to the satellite, a possible inter-satellite path, the satellite to a gateway, and the terrestrial connection to the destination. The propagation component of delay is generally lower for LEO systems than for GEO systems because the signal travels a shorter distance. However, the end-to-end delay also depends on gateway placement, routing, processing, scheduling and retransmissions. Therefore, the lower orbit should not be treated

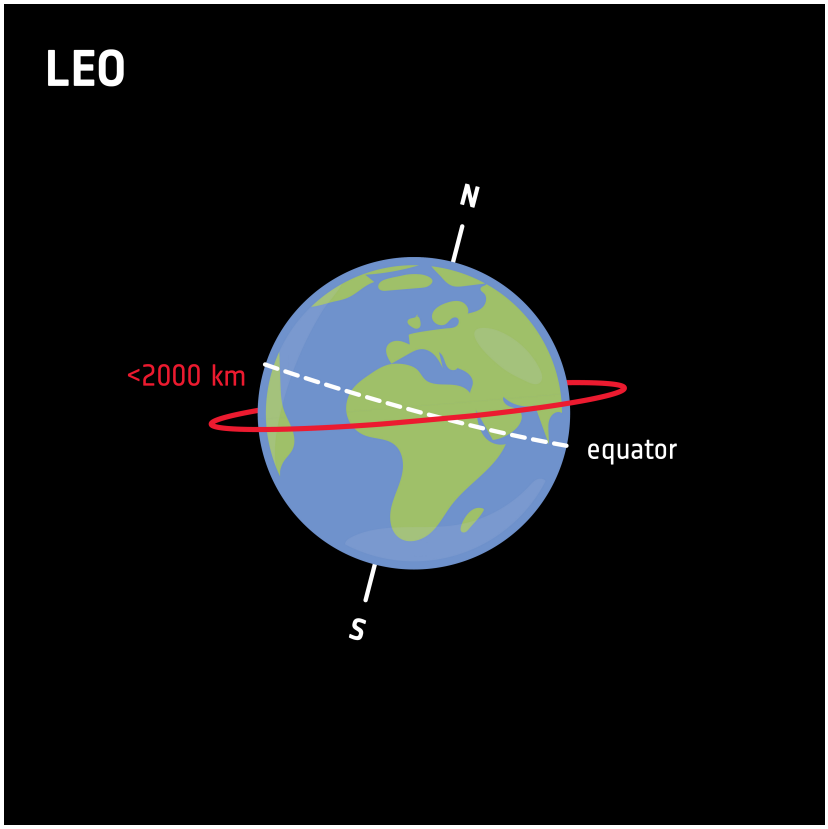


Figure 2: Illustration of low Earth orbit (LEO). The red band indicates the typical LEO altitude range below 2000 km surrounding Earth. Image credit: ESA–L. Boldt-Christmas, reproduced under ESA standard licence for educational and informational use.

as a guarantee of a constant or minimal application-level latency. (Sun, 2014)

LEO links are also affected by the relative motion between the satellite and the user terminal. This motion changes the received signal conditions and produces Doppler shifts that must be compensated by the communication system. Atmospheric conditions, obstructions, antenna visibility, interference and variations in available radio resources can further affect the link. Link quality can consequently vary during an otherwise continuous application session. These variations are important for CBRN systems because a short period of degradation can delay an alert, interrupt a telemetry stream or trigger transport-layer loss recovery.

Satellite networks traditionally use link-layer mechanisms such as error-correcting codes, retransmission and bandwidth allocation to improve the quality and availability of the radio link. These mechanisms can hide some transmission errors from higher layers, but they can also add delay and cause the observed round-trip time to vary. The transport protocol therefore operates over a path whose capacity, delay and loss characteristics may change over time. Measurements of an LEO application should differentiate between the radio-link behavior and the end-to-end behavior observed by the application. (Sun, 2014)

2.2.3 Mobility, Coverage and Handovers

In a terrestrial fixed network, an endpoint can often remain attached to the same access point for the duration of a connection. This assumption does not generally hold for LEO networks. As a satellite moves across the sky, the user terminal may need to transfer its connection to another satellite or gateway in order to maintain service. A handover can involve establishing a new physical link, updating routing information and preserving the reachability of the user terminal. The network may also need to coordinate handovers between the space and ground segments. (Sun, 2014)

A handover does not necessarily cause a complete application outage, but it introduces a period during which the available path can change or packets can be delayed, reordered or lost. The practical effect depends on how the LEO provider implements mobility and on whether the user terminal retains the same network-layer identity. From the perspective of the transport layer, a path change can appear as a temporary increase in round-trip time, a reduction in throughput, packet loss or a connection interruption. These effects make handover behavior a relevant test condition for the resilience evaluation in this thesis.

2.2.4 IP Networking and Transport Implications

LEO satellite networks can be integrated with terrestrial IP networks through gateways and satellites may also contain routing functionality. In a constellation, routing is more complex than in a system consisting of one large-coverage satellite because the connectivity between nodes changes over time. Since satellite positions are predictable, orbit information can be used to anticipate topology changes and update routing decisions. Nevertheless, routing and mobility mechanisms must ensure that packets continue to reach the correct user terminal when its point of attachment changes. (Sun, 2014)

These characteristics create a distinction between link continuity and session continuity. A temporary disruption in one radio or network path does not necessarily mean that the communicating endpoints have become permanently unreachable. For a near-real-time CBRN application, the transport protocol should recover from short disruptions without requiring unnecessary application-level reinitialization, while still exposing enough information to manage stale or delayed data. This is one reason why connection establishment, loss recovery, multiplexing and behavior during path changes are important when selecting and evaluating a transport protocol over LEO satellite internet.

The satellite network also introduces a shared and potentially variable communication resource. Competing traffic, scheduling decisions and gateway capacity can affect the service available to an individual user terminal. Quality of service is therefore not determined only

by the nominal bandwidth of the satellite system. Latency, jitter, packet loss, service availability and the priority of individual data flows are also relevant. For CBRN telemetry, a small and time-sensitive alert may be more important than a larger non-critical payload, so the communication design should consider both the network conditions and the relative importance of the data.

2.2.5 Starlink Satellite Network Characteristics

Starlink is currently the largest commercial LEO satellite constellation with over 7,800 active satellites as of early 2026 and a total authorized fleet of up to 15,000 second-generation (Gen2) spacecraft across multiple orbital shells (Benghe et al., 2026; Federal Communications Commission, 2026). The first-generation constellation operates in four primary shells at inclinations of approximately 43°, 53°, 70°and 97.6° (Benghe et al., 2026). The Gen2 upgrade authorizes additional satellites at lower altitudes ranging from 340 km to 485 km across a wider range of inclinations (Federal Communications Commission, 2026). The constellation employs a bent-pipe architecture where user terminal traffic is relayed through one or more satellites via inter-satellite links before reaching a ground station connected to the broader Internet through a Point of Presence (PoP). Due to the high velocity of LEO satellites moving at approximately 7.6 km/s, user terminals must perform handovers roughly every fifteen seconds as individual satellites enter and exit their field of view.

Performance measurements across Starlink's global coverage reveal significant geographic differences in both latency and throughput (Benghe et al., 2026). Factors contributing to this variation include overhead satellite density which influences handover frequency and routing options, distance to the nearest PoP which determines terrestrial path length after the satellite segment, and atmospheric conditions affecting Ku/Ka-band links. Analysis indicates that geographic position is the dominant factor affecting performance, with latitude alone accounting for up to forty-six percent of latency variation while weather contributes fourteen to fifteen percent depending on location.

Among meteorological variables precipitation has a stronger impact on throughput than on latency reflecting the known sensitivity of Ku/Ka-band signals to rain fade while wind speed contributes more substantially to latency variation potentially through mechanical effects on antenna alignment (Benghe et al., 2026). Satellite density directly influences connectivity quality as regions with higher numbers of overhead satellites benefit from smoother handovers and better routing flexibility. These characteristics make Starlink a representative platform for evaluating transport protocol performance over modern LEO networks but also underscore that end-to-end behavior can vary considerably with location time and environmental conditions.

2.2.6 Challenges Relevant to CBRN Communication

The main advantage of LEO satellite networks for CBRN defense is that they can provide connectivity in locations where terrestrial infrastructure is unavailable, damaged or deliberately disrupted. At the same time, the network has characteristics that complicate the delivery of near-real-time information. The most relevant challenges are variable delay, packet loss, changing throughput, intermittent visibility, handovers, gateway dependence and limited control over the underlying satellite network. These challenges can occur independently or at the same time, which makes average latency alone insufficient for evaluating the communication system.

For this thesis, the relevant performance properties are end-to-end latency, connection establishment time, packet loss, recovery after temporary disconnection and the delivery behavior of high-priority telemetry during network changes. The LEO network is treated as an external communication environment whose detailed radio and constellation control mechanisms are not modified by the proposed artifact. The artifact instead operates at the transport and application boundaries, where it can measure the resulting network behavior and apply transport-level mechanisms to support resilient data delivery. The following section introduces QUIC as the transport protocol selected for this purpose.

2.3 QUIC Protocol

The QUIC transport protocol was designed by Google to solve several problems that had built up in the TLS/TCP stack over the years as the web moved towards encrypted and latency-sensitive traffic (Langley et al., 2017). TCP had been working as the main transport on the Internet for a long time but its deployment had become restrictive. Middleboxes like firewalls and NATs inspect and modify TCP headers as part of their normal operation which makes it hard to deploy even small improvements without co-ordinating changes across OS vendors, middlebox vendors and network operators all at once. Additionally, TCP is usually implemented inside the operating system kernel so modifying it requires OS upgrades that take a long time on both servers and clients. These problems motivated a new design where transport logic can be deployed and improved at application update speed rather than waiting for OS release cycles.

2.3.1 Design Overview

QUIC replaces the traditional HTTPS stack of TCP, TLS and HTTP/2 with one multiplexed encrypted transport that runs over UDP (Langley et al., 2017). Because it operates in user space, QUIC can be shipped as part of applications like web browsers, which allows for rapid iteration through A/B experimentation at Internet scale. Using UDP instead of TCP means

QUIC packets can go through existing middleboxes without special treatment since UDP is allowed on almost all networks today.

One of the main goals of QUIC is to reduce how long it takes to establish a connection. With the old stack a new HTTPS connection needs one round trip for the TCP handshake then two more round trips for TLS before any application data can be sent. QUIC puts together its cryptographic and transport handshakes into one exchange (see Figure 3). On a first connection to a server the client does a one-round-trip (1-RTT) handshake. After a successful connection though the client caches server credentials including a long-term Diffie-Hellman public value and an authenticated source-address token so that later connections to the same server can start with application data already encrypted in the first packet (0-RTT). In Google's production deployment about 88% of all QUIC connections from desktop clients achieved this 0-RTT handshake (Langley et al., 2017). This is especially useful in high-latency environments: while TLS/TCP handshake latency grows linearly with RTT QUIC's average handshake latency stays almost flat because most connections cost zero round trips.

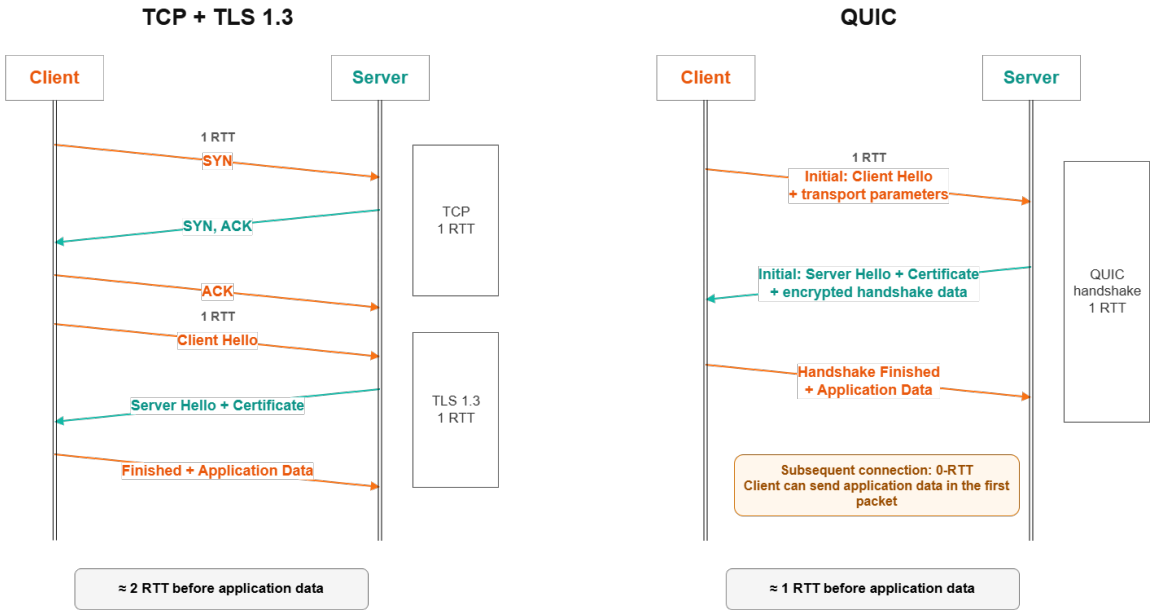


Figure 3: Comparison of connection establishment between TCP+TLS 1.3 and QUIC. TCP requires one round trip for the TCP handshake followed by another for TLS before application data can be exchanged (approximately 2 RTT total). QUIC combines its cryptographic and transport handshakes into a single exchange completing in 1 RTT for the first connection and 0 RTT for subsequent connections to the same server.

2.3.2 Stream Multiplexing and Head-of-Line Blocking

However, the benefits of a faster handshake alone are not enough if the transport still suffers from head-of-line blocking during data transfer. HTTP/2 sends multiple requests over

one TCP connection to save overhead but TCP's sequential bytestream abstraction creates a problem. When a packet gets lost, every stream whose data comes after that packet in the byte stream has to wait until it is retransmitted before it can be delivered to the application. QUIC fixes this by providing lightweight bidirectional streams inside each connection (Langley et al., 2017). Each stream delivers data independently, so losing one UDP packet only affects those streams whose data was carried in that packet. Other streams keep delivering data without interruption.

Streams are identified by numeric IDs allocated separately for client-initiated (odd numbers) and server-initiated (even numbers) streams to avoid collisions. Creating a stream happens implicitly when sending on an unused ID which makes them cheap enough for fine-grained use, such as dedicating one stream per small request-response pair. A FIN bit signals clean stream closure, or either endpoint can cancel an individual stream without tearing down the whole connection.

2.3.3 Loss Recovery and Congestion Control

QUIC improves loss detection over TCP with two design choices (Langley et al., 2017). First, every QUIC packet carries an increasing packet number whether it carries original or retransmitted data. This removes TCP's retransmission ambiguity problem where an acknowledgment cannot tell if it is acknowledging an original transmission or its retransmission, which enables simpler, more accurate loss detection without needing expensive timeouts after lost retransmissions. Second, QUIC acknowledgments explicitly encode the delay between receiving a packet and sending its acknowledgment back. Together with monotonic packet numbers this gives precise RTT estimation that helps both loss detection and delay-sensitive congestion controllers.

QUIC acknowledgments support up to 256 ACK blocks per frame compared to TCP SACK which normally supports three or four selective acknowledgment blocks (Langley et al., 2017). This makes QUIC much more resilient to reordering or burst loss because it can keep more bytes in flight during recovery.

The protocol does not require any specific congestion control algorithm, instead it provides a pluggable interface for experimentation. In Google's deployment both TCP Cubic (the default controller) and its multi-stream variant mulTCP are used depending on whether multiple application streams share one connection (Langley et al., 2017).

2.3.4 Connection Migration

A feature particularly relevant for mobile communication scenarios is QUIC's Connection ID mechanism (Langley et al., 2017). While TCP identifies connections by IP address-port 5-tuple, which causes established connections to break when NAT rebinding happens or when a client changes networks, QUIC identifies connections through an explicit 64-bit Connection ID placed outside encryption cover for routing purposes. This allows endpoints to survive IP address changes transparently: when NAT timeout or rebinding changes visible addresses (common with UDP) or when moving between different networks changes IP assignment entirely, either endpoint can keep using existing connection state identified by Connection ID instead of starting from scratch.

In LEO satellite networks user terminals regularly switch between satellites as they move across overhead orbits, therefore IP-level path changes happen often due to satellite diversity and gateway switching at ground stations. Combined with NAT rebinding which is more aggressive for UDP than for TCP in carrier-grade NAT deployments, these conditions cause regular disruptions to conventional transport connections. The ability of QUIC to maintain connection state across address identity changes reduces the need for full re-establishment after each handover, making it a promising candidate for the dynamic network environment described earlier in this chapter.

2.3.5 Deployment Experience and Relevance to Domain

The effectiveness of these design choices has been validated through large-scale deployment. Langley et al. (Langley et al., 2017) reported that after global deployment on thousands of servers and in Chrome and YouTube clients an estimated 7% of Internet traffic was QUIC by early 2017. Results showed a mean search latency reduction of 8.0% for desktop users and 3.6% for mobile users together with reductions in video rebuffer rates of 18.0% for desktop and 15.3% for mobile playbacks. These improvements were largest at higher RTTs which reflects the fact that saving handshake round trips is more impactful on longer paths. Over 20% of all connections had a minimum RTT exceeding 150 ms which is exactly where satellite links operate (Langley et al., 2017).

For the CBRN communication scenario described in the introduction, first-responder devices operating in unpredictable connectivity conditions can benefit from several of QUIC's properties. Rapid 0-RTT connection establishment allows telemetry and alert delivery immediately upon entering coverage without waiting for a multi-round-trip handshake. Multiplexed streaming prevents a single lost packet containing critical sensor data from blocking less urgent status updates that are sent at the same time on independent streams. Connection-level and stream-level flow control makes sure neither a slow-processing command centre nor a

constrained field device gets overwhelmed by arriving data during degraded network conditions. The Connection ID-based identity means that a temporary interruption caused by satellite handover does not force the application to restart its transport connection, once the link comes back transmission can resume using previously negotiated state.

The following chapter describes the methodological approach used to evaluate the feasibility of applying QUIC in the context of inter-vehicle LEO-based communication through system modelling and simulation in a controlled environment.

3 Research Methodology

This chapter describes the research methodology used in this thesis. The work follows the Design Science Research Methodology (DSRM) framework as described by Peffers et al. (2007). DSRM was selected because the thesis involves both the creation of a software artifact and the evaluation of its performance in a real-world environment which aligns well with the design science approach. The artifact itself was comissioned by an employer with specific requirements for improving the communication capabilities of their existing systems, therefore a practical need was driving the design decisions alongside the research objectives.

3.1 Design Science Research Methodology

Design science research is concerned with creating and evaluating IT artifacts that are intended to solve identified problems (Peffers et al., 2007). It differs from other research frameworks by not only trying to understand reality but also attempting to create something that serves a purpose. The DSRM framework provides a process model for conducting this type of research consisting of six activities: problem identification and motivation, definition of the objectives for a solution, design and development, demonstration, evaluation and communication.

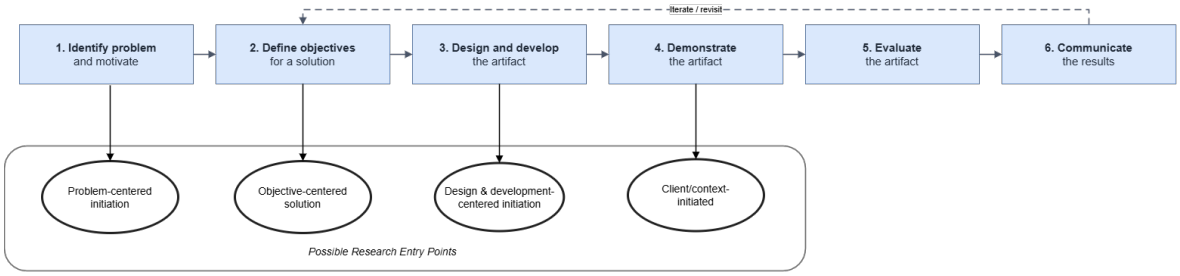


Figure 4: DSRM process model as presented by Peffers et al. (2007). The process includes six activities and four possible entry points depending on how the research is initiated.

The framework allows for different entry points into the process depending on how the research is initiated (Peffers et al., 2007). The problem-centered approach starts from activity 1 when the idea comes from observing a real problem. An objective-centered solution starts from activity 2 when there is an industry or research need that can be addressed by developing an artifact. A design and development-centered approach starts from activity 3 when an artifact already exists or is being built before formalizing the research questions around it. The research proccess begins from the activity associated with the initiator, whereas other activities can be optionally excluded if they pose no relevancy to the research.

In this thesis all three entry points are relevant to some extent. The primary motivation is design and development-centered, because the artifact was originally developed as part of

employment at Observis Oy. A practical need for improved CBRN communication capabilities over satellite networks arose from customers' requirements. Combined access to existing domain knowledge codebases and testing infrastructure shaped what could realistically be built and evaluated.

At the same time there is also a problem-centered motivation behind this work. CBRN defense teams operate in environments where terrestrial communication infrastructure may be unavailable, damaged or deliberately disrupted as discussed in Chapter 1. The limitations of traditional transport protocols like TCP over high-latency lossy satellite links are well documented in literature (Langley et al., 2017), resulting in a clearly identifiable problem that needs solving.

Furthermore, an objective-centered perspective is also applicable because new technologies, such as commercial LEO satellite internet (Starlink) and modern transport protocols (QUIC) have emerged recently, making it possible to achieve better latency and coverage than what was previously feasible with older technology combinations (Benghe et al., 2026). These technological developments created new possibilities that did not exist before, therefore defining objectives around them became meaningful.

This work is at the intersection between industrial practice and academic research, which is reflected by the relevance of the three entry points. This hybrid approach is consistent with how DSRM has been applied across different fields where multiple motivators often coexist within same project (Peffers et al., 2007).

3.2 Research Process

The DSRM framework is followed throughout this thesis by mapping each activity to corresponding chapters:

- **Activity 1 — Problem Identification and Motivation:** Addressed in Chapter 1 where CBRN defense field, LEO satellite networksand QUIC protocol are introduced alongside motivation, context, objectives and scope delimitations.
- **Activity 2 — Objectives for a Solution:** Defined through research questions RQ1 and RQ2, hypotheses H1 and H2, together with concrete objectives O1-O4 outlined in Chapter 1. These objectives describe what a successful outcome looks like.
- **Activity 3 — Design and Development:** Covered in Chapter tbd, describing system architecture protocol design, implementation details and limitations simplifications made during development.
- **Activity 4 — Demonstration:** Conducted through field measurements over Starlink

network using TCP baseline, QUIC implementation and QUIC+ML variant with results presented in Chapter tbd.

- **Activity 5 — Evaluation:** Analysis comparing observed results against defined metrics hypotheses, leading to discussion of findings and validity threats in Chapter tbd .
- **Activity 6 — Communication:** This thesis itself serves as the primary means for the communication of findings to academic audience and field professionals.

A central theme throughout this work is that new technology emerges faster than traditional communication protocols can adapt to changing conditions. This particularly applies to satellite environments, where latency and handovers packet loss vary significantly, depending on location, weather and constellation state. Instead of designing completely new protocols from scratch, this approach investigates whether combining existing technologies such as QUIC and LEO satellite internet can provide acceptable performance for CBRN use cases without requiring modifications of underlying network infrastructure. The theoretical background gathered in Chapter 2 establishes necessary context for understanding why these particular technologies were selected and how they interact with each other.

The evaluation metrics defined later in Chapter tbd reflect what can be practically measured, given the constraints of real-world testing over commercial satellite link. Lower latency is generally considered better, but cannot be measured in isolation as throughput, jitter, processing overhead and recovery time after handover events must be considered together to form a complete picture of transport performance.

3.3 Evaluation Metrics

3.4 Experimental Design

4 Artifact Design

This chapter outlines the system architecture and design decisions across each layer of the artifact. The artifact design is comprised of the following individual parts: high level system design, network topology outline, software application design and raw data protocol design. The various components and systems of the artifact are designed with common design principles and cohesion across the various layers in mind. The deliverables include the system design and a software application referred to as SatLINK.

4.1 High Level Design

A high level design describes the high level objectives of the artifact. Additionally, the design outlines boundaries, links, systems and components comprising the software artifact. The primary design decisions are driven by the technology stack and network provider selection described in Chapter 1. Additionally, the availability of computing hardware, local network providers, internal documentation of CBRN device communication protocols and pre-existing knowledge of CBRN software systems influence the design further. The high level goal of the artifact is to enable and benchmark the bi-directional traffic of telemetry data across public internet infrastructure within the project specific constraints.

4.1.1 Component Overview

The artifact is designed with two distinct computing units in mind. The first unit is a Kuber 212E industrial grade computer, which represents a remote mobile unit. This unit is responsible for running the CBRN data generation and protocol handling components of the artifact. The second unit is a virtual machine hosted on a server grade computer provided by the employer, which represents a command center. The two units are connected over public internet infrastructure using a commercial LEO satellite network (Starlink) as the primary communication link. Terrestrial internet infrastructure is used to connect the central data collection node to the satellite provider's ground gateway as shown on Figure 5.

The SatLINK software application provides the features required by both the mobile unit and command centre nodes. This is achieved by implementing both the server and client functionality, selectable at start-up time via command line arguments. Additionally, the decision to bundle all functionality in a single software package, simplifies distribution and ensures consistency by using the same dependencies, interfaces and logic across both node types.

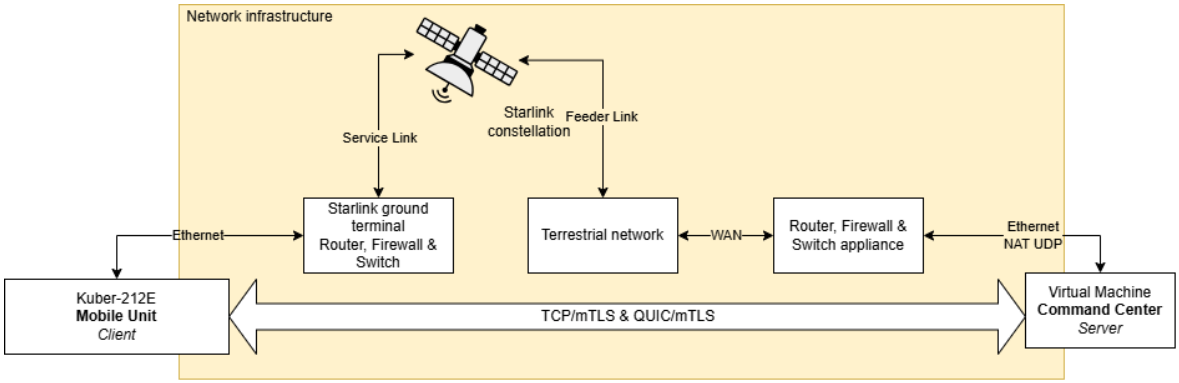


Figure 5: High-level communication link between the remote mobile unit and the command center.

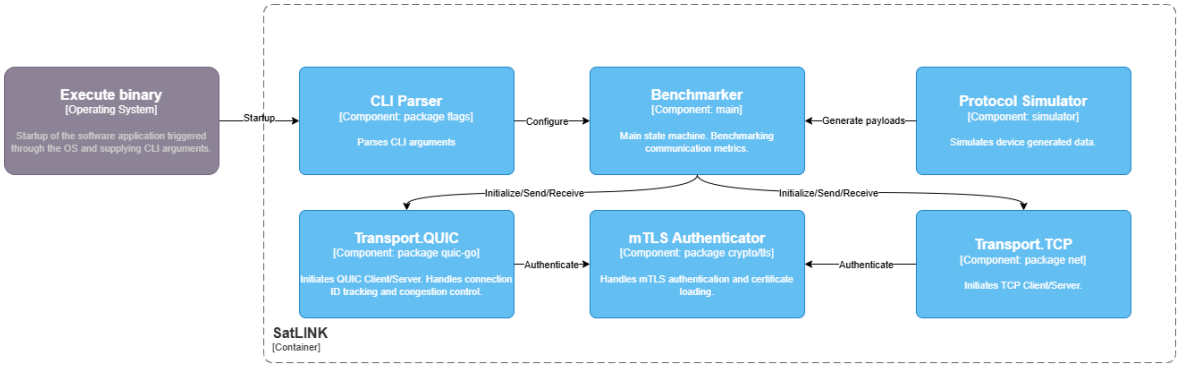


Figure 6: C4 diagram detailing the various components within the software application SatLINK.

4.1.2 Communication Protocol Stack

The transport layer is utilizing two interchangeable protocols: TCP and QUIC. These protocols are chosen to align with the research objectives defined in Chapter 1, allowing direct comparison between the conventional TCP/TLS baseline and the QUIC-based approach. Both protocols operate over the same physical LEO satellite link, as seen in Figure 5, so that the comparison isolates the effect of the transport protocol. The software application communicates using a single protocol at a time that can be selected at start-up time by providing a command line argument. This functionality enables isolated A/B testing of both transport protocols separately, avoiding interference when using both simultaneously.

The application layer uses a raw ASCII-based data protocol designed specifically for this project. The protocol structure is loosely based on the internal communication protocols used by CBRN device manufacturers. This design choice ensures that the simulated payloads are representative of real CBRN system traffic without requiring access to proprietary hardware or vendor-specific interface documentation.

4.1.3 Programming Language and Toolchain

The artifact is implemented in Go (Golang). Go was selected because it offers native or well-supported libraries for the key technical requirements of the project: mutual TLS (mTLS) authentication for secure peer-to-peer communication, QUIC transport with the ability to customize congestion control algorithms, and integration with numerical and machine learning libraries. Furthermore, Go is a production-grade language commonly used at Observis Oy, which simplifies the future maintenance and encorporation of the artifact within Observis Oy's software platform. Additionally, Go features cross-compilation, allowing developers to compile self-contained portable binaries for multiple platforms from a single development environment.

4.2 Protocol Design

4.3 Design Limitations and Simplifications

5 Implementation

5.1 TCP Baseline Implementation

5.2 QUIC Implementation

5.3 Machine Learning Integration

5.4 Benchmarking Tooling

6 Demonstration and Evaluation

6.1 Experimental Setup

6.2 Baseline Measurements

6.3 QUIC Measurements

6.4 Comparative Analysis

6.5 Machine Learning Results

6.6 Discussion

7 Conclusions

7.1 Summary of Contributions

7.2 Limitations

7.3 Future Work

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Appendix 1 Text processing and layout in a thesis

Good text processing skills make writing your final thesis and using this template easier. Therefore, you should ensure you have sufficient basic skills to edit long documents with LaTeX before you start. This involves applying LaTeX code structures, understanding automatic referencing and knowing how to divide your text into sections.

The essential thing is to understand the following basics of LaTeX:

- LaTeX needs a compiler. Compiler can be stand-alone in the computer or cloud-based, such as Overleaf
- LaTeX will remove all the unnecessary spaces and linebreaks. When you want to start a new paragraph, make one empty line into your code, LaTeX will start the paragraph.
- Do not enter any numbering manually. LaTeX has efficient automatized tools for this. Section, page, figure, and table numbering are automatically applied in this template. They keep the numbers in the right order even if you modify, add or remove content.
- Do not add hyphenation at the end of a line manually. LaTeX automatic hyphenation tool can be used in this template. It should work with both English and Finnish languages.

Line spacing, font, margins, alignment, page numbering and headings

Official layout guidelines state that the line spacing should be 1.5, except for the abstract and possible direct citations, where the spacing is 1. You can choose from two fonts: Times New Roman (12 pt) or Arial (11 pt). This template has been written in Adobe Times Roman 12 pt.

Leave the following margins:

- top and left 35 mm.
- bottom and right 20 mm.

The page number of the title page is 1, but the page numbering should not be visible before the first page of the table of contents. Place the page numbers at the top, either centred or in the right-hand corner. Page numbering ends on the final page of the reference list: appendices do not have page numbers unless the appendix is multiple pages. For a single-page appendix, use the command `\thispagestyle{empty}` to remove the page number.

Always use section styles in your thesis (`\section{}`, `\subsection{}`, `\subsubsection{}`).

Always place section headings (`\section{}`) on a new page. Apply `\clearpage` command before the section heading starts the section from the new page. If you add large figures or tables, remember to check that the empty space after the figure or image covers no more than 20% of the page.

The heading numbering starts from the introduction and continues consecutively. Do not place a period after the heading number. Do not number the heading of the reference list at the end of the thesis.

The thesis should include no more than three heading levels, and the headings should progress logically. If you need even more detailed subheadings, do not number them; leave them out of the table of contents. Concise headings that describe the text sufficiently are the best. You can use question marks or exclamation marks but do not add a period if the heading is a regular sentence.

A heading cannot follow a heading. Always write something between them. For instance, there must be text between headings 1 and 1.1 with two or more sentences.

Lists are a good way to express things clearly. Use the same type of bullet or symbol in lists throughout your thesis. A section should never end in a list. There should always be two or three sentences after a list.

Appendix 2 References

The text must include references to the sources you use. LUT University applies the Harvard referencing style, also called the author-date style, with in-text referencing and a detailed reference list at the end.

The purpose of a reference list is to provide sufficient information on a source used in the study, allowing the reader to consult the original source for further information. The reference enables the reader to find detailed information on the source easily in the list of references. You should refer to the original and most recent sources. If no new studies have been published on the topic in question, also older ones may be used.

Referring to a source means that you explain the contents of the source material in your own words. Direct citations, on the other hand, are placed inside quotes “Quote”. Plagiarism or using another person’s original material without appropriate referencing is not allowed.

Referencing technique

In the Harvard system, the citation is placed in parentheses directly in the text to indicate the passage that has been cited from another source. Place the citation before the period that ends the sentence when it refers only to the sentence in question (Geldart, 1985). If you are referring to more than that one sentence, introduce the source you are summarizing or paraphrasing at the beginning of the paragraph. Then, refer back to the source when needed to ensure your reader understands you are still using the same source. In-text references, specifying page numbers, are not used unless a specific part of the text is directly referred to or it is a direct quotation. Please note that the author does not always have to be an individual but can also be, for example, an organization. If the publication’s author does not appear in the source, the name of the publication is referred to instead of the author’s name.

Author’s name cited in the text (direct)

Direct citation is used when the author’s name(s) is a part of the sentence structure, with the year in parentheses. The following is obtained by using `\citet{key}` command.

> Allen & Tildesley (1987) claims that . . .

More than three authors for a work

> Chowdhury et al. (2017) in their work . . .

On some special occasions, all the authors (more than three) can be cited, but this form should be considered only at the *special* case by using `\citet*{key}` command

> Bizon, Shattuck, Swift, McCormick & Swinney (1998) showed ...

Author's name not cited in the text (indirect)

Indirect citation is used when the author's name(s) is not mentioned directly in the sentence structure. The indirect citation will appear at the relevant point in the sentence or at the end of the sentence in parentheses. The following is obtained by using `\citep{key}` command.

> Earlier research claims that ... (Savage, 1992).

Two authors with postnote

> At the theoretical approaches of ... (Allen & Tildesley, 1987, p. 102).

More than three authors for a work

> ..., (Veleva et al., 2015).

Combined citation with three entries using the command `\citep{key1,key2,key3}`

> According to previous studies, the presented conclusions are consistent (Miller, 1997; Dalton, Petri & Pontuale, 2005; Allen & Tildesley, 1987).

In many situations, it is justified to combine the content of several sources, and then it is recommended to use content-specific indirect references. For example, in the following way:

> "In literature we can find challenges relating to a wide variety of issues including environmental sustainability (Dagiliute & Liobikiene, 2015), business needs (Veleva et al., 2015), food production (Chowdhury et al., 2017), urban development (Joss, 2010; Swapan, 2016; Freeman, McMahon & Godfrey, 2017; Waite et al., 2017), and education (Zidansek, 2007; Biberhofer & Rammel, 2017)." (Talikka, 2018)

A wide variety of sources can be used in the text. In addition to the above, Master's theses (Hyvärinen, 2019), websites (NASA, 2015), interviews (Kottarainen, 2007, interview 6.6.2007), technical reports (Laine, Leppänen & Nurmi, 2009), booklets (Swetla, 2015), manuals (R Core Team, 2018), proceedings (Stepney & Verlan, 2018), standards (ISO 9004, 2000), patents (Virtamo & Aalto, 1996) or AI-language models (OpenAI, 2023) can be used.

All the bibliography information is given in the References. If you refer several works published by the same author in the same year, add lowercase letters (a, b, c. . .) after the publication year to distinguish the sources. Use the same alphabetical notation also in the list of references. There are several referencing and citation styles. It is essential to use the same style throughout the thesis consistently. Examples and detailed instructions on referencing:

- LUT Academic Library's instructions on how to cite electronic documents
- Aalto University citation guide
- Harvard referencing, University of Sheffield

Appendix 3 Tables, figures, equations, numbers, symbols, and abbreviations

It is a good idea to illustrate your text with figures and tables. Figures and tables must have captions and consecutive numbering. The captions of tables are placed above the table, and those of figures are below the figure. Refer to the figures and tables in the text body, preferably before you introduce them, and align them with the text body. In many theses, the alignment of figures and tables has been centered and without wrapping text around them, but the author can select the formatting options as long as they are coherent in the whole work. LaTeX tries to place all floating objects (figures and tables) at the top of the page or in the best place and to avoid multiple floating objects on the same page. Because of this, for several floating objects in series, their position can be shifted several pages forward. It is also possible for floating objects to move to the next section. Because of this, several consecutive floating objects should be avoided, and enough text should be added between the floating objects. In this template `\usepackage[section]{placeins}` setting is used to force floating objects to stay in the appearing section. You can also use `\FloatBarrier` command to set a position limit for floating objects.

Remember to add alt text (alternative text) to your figures and tables to ensure accessibility. Alt text is read by a designated reader and can be viewed even when the image cannot be displayed on the page. In LaTeX the alt text has to be set manually and you should make sure it describes the object sufficiently and understandably. You can add the alt text by using the command `\pdftooltip{Object}{Alt-text}`.

Tables

Give your table number and caption, and refer to them in the text. Place the caption above the table, name its columns, and mention the units applied, as in Table 1 below. Avoid empty columns or rows. The recommended font size is 10.

Table 1: Sensor measurements.

Voltage U [V]	Pressure p [Pa]
0.984	0
2.252	150
2.772	300
3.181	450
3.615	600
3.817	750
4.088	900

Figures, charts, graphic elements

Images help illustrate your text. The text should contain a reference to every figure. Number your figures and place a caption underneath – not inside the figure.

You should use a software program like Excel or Matlab to draw charts. Charts should be clear and easy to understand. Use a white background. A background grid is allowed if it does not make the figure difficult to interpret. Variables and measurement points should be clearly visible. Name the axes and their units. An example script to produce a Figure 7 with Matlab is presented in Appendix 6. To maintain full accessibility of the document, you should use embedded fonts in the vector figures. The example script will not produce a figure with embedded fonts, and the jpg or png formats would be better with respect to document accessibility. However, the figure quality is reduced.

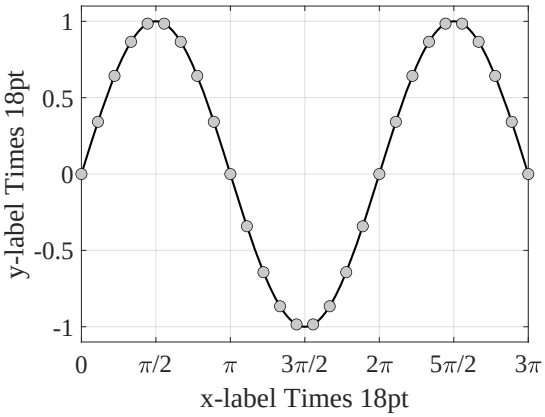


Figure 7: Illustration of Matlab example figure.

Create as many of the figures yourself as you can. Use the same font as in the text body and equations. Try to maintain the font size in the figure as close as possible to the main text font size. If you use images created by someone else, remember to cite them correctly. Remember that images are copyrighted works, the use of which must always be authorized by the author. Captions need to be in the same language as the text body.

Figure 8 shows a gas fermenter in a laboratory configuration. The image is positioned at the beginning of the page, and with the command `\FloatBarrier`, the following text is forced to appear after the image.

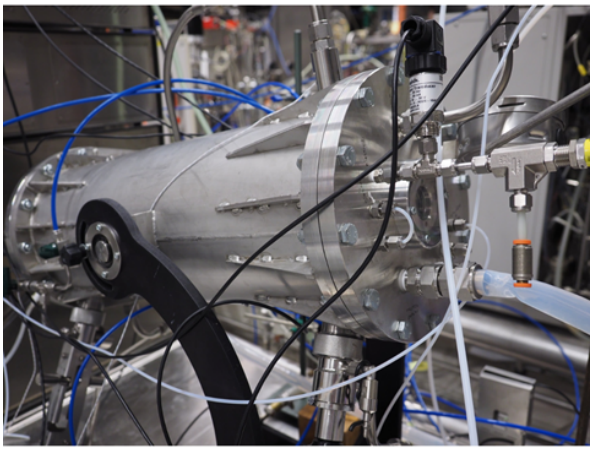


Figure 8: Gas fermentor (VTT 2020, LUT image bank).

Do not end a paragraph in a figure or table. Add text underneath, such as comments on the figure. Large figures, tables, long equations, and other supporting material can be appended.

Numbers, symbols and equations

Numbers in the text are usually approximations. Their accuracy depends on the observational error. Include only significant figures in the results. Interim results should include at least two figures more to avoid round-off errors. Present large and small figures in powers of ten 10^n, where n should preferably be divisible by three.

Equations and other mathematical expressions must consist of standardised symbols if one exists. You may use other symbols only if there are no applicable standardized or established ones. Variables and constants are denoted using *slanted style*, vectors are denoted using **bold regular style**, and abbreviations and dimensionless numbers are denoted using regular style. See the SI-unit style guide.

Explain the symbols in an equation when you use them for the first time. Write each equation clearly on its own line and indent it. Number your equations consecutively or by paragraphs so that the number is in parentheses on the right side of the equation and aligned to the right. You can refer to an equation only after you have presented it, with certain exceptions, such as if the object you are referring to is far ahead. As a simple example, the equation of state of an ideal gas can be used

$$pv = RT, \tag{1}$$

where p is pressure [Pa], v is specific volume [m^3/kg], R is the gas constant [J/(kgK)] and T is temperature [K].

The SI-unit style should be followed. Here are some common guidelines for using variables, constants, and units correctly:

- Write scalars in italics and vectors in bold, not in italics. Don't write dimensionless quantities in italics.
- Don't use $*$ multiplier in your equations. Present the variables to be multiplied in the equations in consecutive order, placing the numerical scalar value first, without any multiplier sign. Exceptions: for 1.452×10^6 or when you place numerical values into the equations in the example calculations, use `\times` command for a multiplier. In the latter case, remember to show the units as well.
- Use only single-character variables to avoid mistakes with previous point. You may use subscripts to separate variables.
- Write subscripts upright unless there is a need to italicise them (they are variables). Write abbreviated subscripts and numerals e.g., as follows: $\Delta\sigma_{\mathrm{w}}$, σ_1, or σ_{min}. For instance, in the summation $\sum_i^\infty x_i$, the subscript i must be italicised because it represents a variable.
- If you wish to express a change in, e.g. pressure Δp, write Δ in a regular font. In some cases, Δ may also be variable and should be italicised. $\pi = 3.14159$ is the ratio of a circle's circumference to its diameter. π may be the pressure ratio.
- Do not italicise mathematical operators such as $\sin(x)$, $\cos(x)$, $\log(y)$, or $\ln(y)$.
- Distinguish absolute values as follows: "variable_=_value_unit", with the exception of a percentage sign after a numeral, e.g. $a = 5.2$ mm, $\gamma = 97.7\%$.
- Use a decimal point "." in accordance with international standards. In contrast, a decimal comma is used in theses written in Finnish. This also applies to figures and tables.

List of symbols and abbreviations

List symbols and abbreviations and their definitions that are not common knowledge separately on their own page before the table of contents. Divide them into groups: Latin characters, Greek characters, subscripts, superscripts, abbreviations, and finally, dimensionless numbers. Give the page the heading Symbols if there are no abbreviations or Abbreviations if there are no symbols.

When you use a symbol or abbreviation in the text body for the first time, introduce it to the reader, for example, as follows: "The concept design for manufacturing and assembly (DFMA) is…". After this, you can use only the abbreviation, and the reader can verify its

meaning from the abbreviation list. Do not add concepts to the list of symbols and abbreviations you do not mention in your text. Don't include standard, well-known elements or species (e.g., oxygen O_2 or carbon dioxide CO_2) definitions in the abbreviations.

Appendix 4 Appendices to the thesis

Appendices may include, e.g., interview questions, survey forms, or other content relevant to the work but not necessary to include in the text body. All appendices are public and should not contain any confidential information.

In your text body, refer to the appendices by adding their title in parentheses (Appendix 1) where relevant. Give all appendices a title based on their content and list them in the table of contents in the order in which they are referred to in the thesis.

Single-page appendices do not require page numbering. Multiple-page appendices do.

Appendix 5 Publishing the thesis

LUT's degree regulations state that Bachelor's and Master's theses are public documents. They are published in the LUTPub repository, and related instructions are available on the library website.

Together with the first examiner, make sure that the commissioner of your thesis is aware of the publicity requirements from the very beginning of the discussions. The student writing the thesis and the thesis commissioner are responsible for ensuring that there is no confidential material in the thesis. The background material of the thesis may contain confidential content. The background material is not public.

All theses in LUTPub must fulfill accessibility requirements. Your text must be as legible as possible to readers. Remember:

- to add PDF meta-data at the beginning of main.tex according to Figure 9 with your own information.
- to use LaTeX commands for headings,
- embed hyperlinks into your text or a description of the linked content; do not include URL addresses,
- add/check the alt text of your figures and tables describing briefly the main content in writing.
- to refer to each figure and table in the text, and verbally describe in the text the key content presented in the figures and tables.

```
%%%%%%%%%%%%%%%%%%%%
%% Give first the meta-data for the PDF/A
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%% IMPORTANT!!!!                                                          %
%% Mandatory data, update with your own data                              %
%% This data will appear in the pdf file properties                       %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
\begin{filecontents*}[overwrite]{\jobname.xmpdata}
\Title{Title of the work}
\Author{Author_firstname Author_lastname}
\Language{en-GB}
\Subject{The abstract or short description.}
\Keywords{template\sep thesis\sep LaTeX}
\end{filecontents*}
```

Figure 9: Example to define PDF meta-data in latex-system.

At the moment (09/2023), LaTeX text system and its add-ons (packages) can only produce pdf files according to accessibility criteria to a limited extent. Using the LaTeX system, it is not possible to produce a final product that meets all accessibility criteria and works universally.

Due to deficiencies, the automatic read-out of the PDF file may have problems, especially with images, tables, and equations. For this reason, it is especially important that the contents of floating objects (images, tables, and equations) are described as clearly as possible in the body text.

Appendix 6 Matlab script for example figure

```
%% This script will generate an example figure and it will export the
    figure in jpg, png and pdf format
% Jouni Ritvanen, 2023

close all
clear

fs=18;                % Font size
fc=[0.79 0.79 0.79]; % marker face color (light grey)

x=0:pi/9:3*pi;   % creating x-vector

figure  % create figure
fplot(@sin,[0,3*pi],'k','LineWidth',1.5)    % plotting sin(x) function
hold on % this allows to add next plot into the figure
grid on % adding grid
plot(x,sin(x),'o',...          % plotting discrete sin(x) points
    'MarkerSize',8,...         % setting marker size to 8
    'MarkerEdgeColor','k',... % setting marker edge color to black
    MarkerFaceColor=fc)        % setting marker face color to fc'

set(gca,'fontname','times')   % setting font to Times
set(gca,'fontsize',fs)        % setting font size to fs
ylim([-1.1 1.1])              % setting y-axes limits
xlabel('x-label Times 18pt')  % setting x-axes label
ylabel('y-label Times 18pt')  % setting y-axes label
xticks(0:pi/2:3*pi)           % setting x-ticks every pi/2
% setting x-ticks labels
xticklabels({'0','\pi/2','\pi','3\pi/2','2\pi','5\pi/2','3\pi'})
%% this section will adjust the paper size to correspond the figure size
% i.e. the output will be the size of Matlab figure, not e.g. A4
pps=get(gcf,'paperposition');   % gets figure size and position
set(gcf,'papersize',pps(3:4)); % sets paper size to correspond figure
    size
set(gcf,'paperposition',[0 0 pps(3:4)]);    % sets figure to fill paper
%% this section will save the figure in jpeg and pdf formats
print -djpeg -r600 example_name % output will in resolution of 600
print -dpng -r600 example_name  % png resoluutiolla 600
print -dpdf example_name        % vector pdf output
```

Appendix 7 Bibliography entries in .bib file

```
%% Reference examples

% Article, two authors
@article{DAGILIUTE,
title           = {University contributions to environmental
    sustainability: challenges and opportunities from the Lithuanian case
    },
journal         = {Journal of Cleaner Production},
volume          = {108},
pages           = {891-899},
year            = {2015},
doi             = {10.1016/j.jclepro.2015.07.015},
author          = {Renata Dagiliute and Genovaite Liobikiene}}
% Article, five authors
@article{VELEVA,
title           = {Understanding and addressing business needs and
    sustainability challenges: lessons from Devens eco-industrial park},
journal         = {Journal of Cleaner Production},
volume          = {87},
pages           = {375-384},
year            = {2015},
doi             = {10.1016/j.jclepro.2014.09.014},
author          = {Vesela Veleva and Svetlana Todorova and Peter Lowitt
    and Neil Angus and Dona Neely}}
% Article, four authors
@article{CHOWDHURY,
title           = {Key sustainability challenges for the global
    phosphorus resource, their implications for global food security, and
    options for mitigation},
journal         = {Journal of Cleaner Production},
volume          = {140},
pages           = {945-963},
year            = {2017},
doi             = {10.1016/j.jclepro.2016.07.012},
author          = {Rubel Biswas Chowdhury and Graham A. Moore and Anthony
     J. Weatherley and Meenakshi Arora}}
% Article, one author
@article{swapan,
author          = {Abu Yousuf Swapan},
journal         = {Journal of Urban and environmental Engineering},
number          = {2},
pages           = {221--232},
title           = {Holistic strategy for urban design: {A} microclimatic
    concern to sustainability},
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volume          = {10},
doi             = {10.4090/juee.2016.v10n2.221232},
year            = {2016}}
% Article, three authors
@article{freeman,
author          = {Rachel Freeman and Chris McMahon and Patrick Godfrey},
title           = {An exploration of the potential for re-distributed
   manufacturing to contribute to a sustainable, resilient city},
journal         = {International Journal of Sustainable Engineering},
volume          = {10},
number          = {4-5},
pages           = {260-271},
year            = {2017},
doi             = {10.1080/19397038.2017.1318969}}
% Article, six authors
@article{WAITE,
title           = {Global trends in urban electricity demands for cooling
    and heating},
journal         = {Energy},
volume          = {127},
pages           = {786-802},
year            = {2017},
doi             = {10.1016/j.energy.2017.03.095},
author          = {Michael Waite and Elliot Cohen and Henri Torbey and
   Michael Piccirilli and Yu Tian and Vijay Modi}}
% Article, one author
@article{ZIDANSEK,
title           = {Sustainable development and happiness in nations},
journal         = {Energy},
volume          = {32},
number          = {6},
pages           = {891-897},
year            = {2007},
doi             = {10.1016/j.energy.2006.09.016},
author          = {Aleksander Zidansek}}
% Article, two authors
@article{Biberhofer,
title           = {Transdisciplinary learning and teaching as answers to
   urban sustainability challenges},
journal         = {International Journal of Sustainability in Higher
   Education},
volume          = {18},
number          = {1},
pages           = {63-83},
year            = {2017},
doi             = {10.1108/IJSHE-04-2015-0078},
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author          = {Petra Biberhofer and Christian Rammel}}
% Article, five authors
@ARTICLE{Biz98,
author          = {C. Bizon and M. D. Shattuck and J. B. Swift and W. D.
    McCormick and H. L. Swinney},
title           = {Patterns in 3{D} Vertically Oscillated Granular Layers
    : Simulation and Experiment},
journal         = {Physical Review Letters},
year            = {1998},
volume          = {80},
pages           = {57--60},
number          = {1},
doi             = {10.1103/PhysRevLett.80.57}}
% Book
@BOOK{All87,
title           = {Computer Simulation of Liquids},
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author          = {M. P. Allen and D. J. Tildesley},
address         = {New York},
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@INBOOK{CitekeyInbook2,
author          = {D. Geldart},
title           = {Elutriation},
editor          = {J. F. Davidson and R. Clift and D. Harrison},
booktitle       = {Fluidization},
edition         = {2nd edn.},
year            = {1985},
publisher       = {Academic Press},
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ISBN            = {0-12-205552-7},
pages           = {383--412},
chapter         = {11}}
% On book
@INBOOK{Joss,
author          = {S. Joss},
title           = {Eco-cities: a global survey 2009},
editor          = {C.A. Brebbia and S. Hernandez and E. Tiezzi},
booktitle       = {The Sustainable City VI: Urban Regeneration and
    Sustainability},
year            = {2010},
publisher       = {WIT Press},
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@INCOLLECTION{Sav92,
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   materials; spatiotemporal coherence and 1/f noise},
booktitle       = {Physics of Granular Media},
publisher       = {Nova Science Publishers},
year            = {1992},
editor          = {D. Bideau and J. A. Dodds},
series          = {Les Houches Series}}
% In proceedings
@INPROCEEDINGS{Dal05,
author          = {F. Dalton and A. Petri and G. Pontuale},
title           = {Stress fluctuations and the solid/fluid transition in
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booktitle       = {Powders and Grains},
year            = {2005},
editor          = {R. García-Rojo and H.J. Herrmann and S. McNamara},
address         = {Stuttgart, Germany, Jun. 18-22, 2005},
pages           = {353--355}}
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@INPROCEEDINGS{Gol98,
author          = {I. Goldhirsch},
title           = {Kinetics and Dynamics of Rapid Granular Flows},
booktitle       = {Physics of Dry Granular Media},
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volume          = {350},
series          = {NATO ASI Series E},
pages           = {371--400},
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% PhD thesis
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title           = {Stress fluctuations in continuously sheared granular
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school          = {Duke University},
address         = {Durham, NC},
year            = {1997}}
% DSc thesis
@PHDTHESIS{Talikka,
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title           = {Recognizing required changes to higher education
   engineering programs' information literacy education as a consequence
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school          = {Lappeenranta University of Technology},
address         = {Lappeenranta, Finland},
type            = {Doctoral thesis},
year            = {2018},
URL             = {https://urn.fi/URN:ISBN:978-952-335-238-4}}
% Master's thesis
@THESIS{Hyv19,
author          = {Joonas Hyvärinen},
title           = {Techno-economic evaluation of carbon capture
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school          = {Lappeenranta-Lahti University of Technology LUT},
address         = {Lappeenranta, Finland},
type            = {Master's thesis},
year            = {2019},
URL             = {https://urn.fi/URN:NBN:fi-fe2019100731528}}
% Website
@MISC{CitekeyMisc,
title           = {Pluto: The 'Other' Red Planet},
author          = {NASA},
howpublished    = {[Website]},
URL             = {https://www.nasa.gov/nh/pluto-the-other-red-planet},
year            = 2015,
note            = {[Cited: 2023-03-06]}}
% Technical report
@TECHREPORT{Lai09,
author          = {J. Laine and T. Leppänen and O. Nurmi},
title           = {The influence of special effects on the perception of
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institution     = {VTT Technical Research Center of Finland},
year            = {2009},
type            = {Research report},
series          = {VTT-R-04966-09},
address         = {Espoo}}
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   author's possession}}
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@booklet{CitekeyBooklet,
title           = {Canoe tours in {S}weden},
author          = {Maria Swetla},
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year            = {2015}}
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% Manual
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title           = {{R}: A Language and Environment for Statistical
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author          = {{R Core Team}},
organization    = {R Foundation for Statistical Computing},
address         = {Vienna, Austria},
year            = {2018}}
% Proceedings
@proceedings{CitekeyProceedings,
editor          = {Susan Stepney and Sergey Verlan},
title           = {Proceedings of the 17th International Conference on
    Computation and Natural Computation, Fontainebleau, France},
series          = {Lecture Notes in Computer Science},
volume          = {10867},
publisher       = {Springer},
address         = {Cham, Switzerland},
year            = {2018}}
% Standard
@TECHREPORT{standardi,
author          = {{ISO 9004}},
title           = {Quality magement systems -- Guidelines for performance
    improvements},
institution     = {International Organization for Standardization},
year            = {2000},
type            = {Standard},
address         = {Geneve, Switzerland}}
% Patent
@MANUAL{US5503249,
type            = {Patent, US-5503249},
author          = {J. Virtamo and S. Aalto},
organization    = {Kone Elevator, SW Baar},
title           = {Procedure for controlling an elevator group},
year = {1996}}
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type            = {[Large language model]},
title           = {ChatGPT, Mar 14 version},
author          = {OpenAI},
URL             = {https://chat.openai.com/chat},
year            = {2023}}
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@article{Otrisal2014,
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  author  = {Pavel Ot\v{r}\'{\i}sal},
  title   = {The Concept of {CBRN} Defence and Possibilities of its
     Application in the {Czech Armed Forces}},
  journal = {Journal of Defense Resources Management},
  year    = {2014},
  volume  = {4},
  number  = {2},
  pages   = {1--4},
  url     = {https://www.longdom.org/open-access/the-concept-of-cbrn-
     defence-and-possibilities-of-its-application-in-theczech-armed-
     forces-21873.html}
}

@manual{JESIP2016,
  title        = {Responding to a {CBRN(e)} event: joint operating
     principles for the emergency services},
  author       = {{Joint Emergency Services Interoperability Programme (
     JESIP)}},
  organization = {HM Government},
  year         = {2016},
  note         = {Available at: \url{https://www.gov.uk/government/
     publications/counter-terrorism-strategy-contest} (Accessed: 13
     November 2019)},
  address      = {United Kingdom}
}

@manual{MOHMalaysia2025,
  title        = {Chemical, Biological, Radiological, Nuclear, and
     Explosives Management Guidelines},
  author       = {{Ministry of Health Malaysia}},
  organization = {Disaster, Outbreak, Crisis, and Emergency Management
     Sector, Preparedness, Surveillance, and Response Section, Disease
     Control Division},
  year         = {2025},
  address      = {Putrajaya, Malaysia},
  isbn         = {978-967-25827-9-3},
  edition      = {First Printing}
}

@inproceedings{Marques2025,
  author    = {Mario Monteiro Marques and Filipe Duarte and Victor Lobo},
  title     = {{CBRN} Emergency Management System},
  booktitle = {Developments and Advances in Defense and Security},
  year      = {2025},
  pages     = {319--330},
  publisher = {Springer},
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  note      = {Chapter 26. First Online: 31 March 2025}
}

@misc{EnvironicsEnviScreenX,
  author       = {{Environics Oy}},
  title        = {{EnviScreenX} {CBRN} System Software},
  howpublished = {Official Product Documentation},
  year         = {2026},
  url          = {https://www.environics.fi/cbrn-products/enviscreenx/},
  note         = {Accessed: 4 July 2026},
  address      = {Mikkeli, Finland}
}

@manual{BruhnNewTechFrontline,
  title        = {{CBRNe} {Frontline}: Technical Product Specification
     and Data Sheet},
  author       = {{Bruhn NewTech}},
  organization = {Distributed by BMD S.p.A. Defense and Security Systems
     },
  year         = {2021},
  url          = {https://www.bmdspa.it/docs/511/bruhn-newtech-cbrne-
     frontline.pdf},
  note         = {Accessed: 4 July 2026}
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@misc{ObservisObsasLink,
  author       = {{Observis Oy}},
  title        = {{ObSAS LINK} --- Integrated {CBRN} Sensor Networking
     and Communications Framework},
  howpublished = {Official Technical Product Overview},
  year         = {2026},
  url          = {https://www.observis.fi/en/obsas-link/},
  note         = {Accessed: 4 July 2026},
  address      = {Mikkeli, Finland}
}

@book{Sun2014,
  author    = {Zhili Sun},
  title     = {Satellite Networking: Principles and Protocols},
  publisher = {John Wiley \& Sons, Incorporated},
  year      = {2014},
  address   = {Chichester, UK},
  url       = {https://ebookcentral.proquest.com/lib/lut/detail.action?
     docID=7104144},
  note      = {Accessed via ProQuest Ebook Central for LUT University}
}
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@article{DuanTong2021SSNC,
keywords = {Starlink ; Aerospace electronics ; co-simulation ;
    Communication ; Communication network ; Communications equipment ;
    Communications systems ; cyber-physical power system ; Cyber-
    physical systems ; Delays ; Extraterrestrial measurements ; free
    space ; Infrastructure ; Internet ; Internet access ; Phasor
    measurement units ; power system simulation ; Power systems ;
    Satellite constellations ; Satellites ; space network ; Space
    vehicles},
language = {eng},
number = {4},
pages = {3673-3675},
publisher = {IEEE},
title = {Starlink Space Network-Enhanced Cyber-Physical Power System},
volume = {12},
year = {2021},
abstract = {The information and communication technology (ICT) is
    increasingly involved in the measurement, operation, control and
    protection of the cyber-physical power system (CPPS). As one of the
    most promising communication infrastructure projects, Starlink is a
    satellite Internet constellation being constructed by SpaceX for
    providing global Internet access, which can benefit the network
    service supply for the communication-enabled power system equipment in
     locations where the network access is unreliable, expensive, or
    completely unavailable. In this letter, the future applications of the
     Starlink space network in CPPSs are explored: the communication
    infrastructure and transmission parameters are discussed, and the
    corresponding test case was emulated in real-time on the heterogeneous
     co-emulation platform to validate the proposed concept of space
    network enhanced cyber-physical power system.},
author = {Duan, Tong and Dinavahi, Venkata},
address = {Piscataway},
copyright = {Copyright The Institute of Electrical and Electronics
    Engineers, Inc. (IEEE) 2021},
issn = {1949-3053},
journal = {IEEE transactions on smart grid},
}

@article{AbelsJoscha2024Piig,
keywords = {Business ; Capital ; Companies ; Cooperation ; Decision
    making ; Geopolitics ; Globalization ; Independence ;
    Infrastructure ; International cooperation ; Multinational
    corporations ; Ownership ; Politics ; Transnationalism},
language = {eng},
number = {4},
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pages = {842–866},
publisher = {SAGE Publications},
title = {Private infrastructure in geopolitical conflicts: the case of Starlink and the war in Ukraine},
volume = {30},
year = {2024},
author = {Abels, Joscha},
address = {London, England},
copyright = {The Author(s) 2024},
issn = {1354–0661},
journal = {European journal of international relations},
}

@inproceedings{KooshaBehzad2025CAoR,
keywords = {Costs ; Earth ; Environmental monitoring ; Internet of Things ; Low earth orbit satellites ; Satellite constellations ; Satellites ; Space vehicles ; Telecommunications ; Timing},
language = {eng},
pages = {1–12},
publisher = {IEEE},
title = {Comprehensive Analysis of Recent LEO Satellite Constellations: Capabilities and Innovative Trends},
year = {2025},
author = {Koosha, Behzad and Madani, Pooria and Ardakani, Mansoor Dashti},
booktitle = {Proceedings - IEEE Aerospace Conference},
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@inproceedings{LangleyAdam2017TQTP,
keywords = {Networks -- Network protocols -- Cross–layer protocols ; Networks -- Network protocols -- Network protocol design ; Networks -- Network protocols -- Transport protocols},
language = {eng},
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publisher = {ACM},
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author = {Langley, Adam and Riddoch, Alistair and Wilk, Alyssa and Vicente, Antonio and Krasic, Charles and Zhang, Dan and Yang, Fan and Kouranov, Fedor and Swett, Ian and Iyengar, Janardhan and Bailey, Jeff and Dorfman, Jeremy and Roskind, Jim and Kulik, Joanna and Westin, Patrik and Tenneti, Raman and Shade, Robbie and Hamilton, Ryan and Vasiliev, Victor and Chang, Wan–Teh and Shi, Zhongyi},

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booktitle = {Proceedings of the Conference of the ACM Special Interest
    Group on Data Communication},
copyright = {2017 Owner/Author},
isbn = {1450346537},
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author = {Peffers, Ken and Tuunanen, Tuure and Rothenberger, Marcus and
    Chatterjee, S.},
year = {2007},
month = {01},
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}

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abstract = {PROTECT\texttrademark is an early-warning crisis management
    system for chemical attacks in high-threat infrastructures such as
    subways, transportation terminals, buildings, and airports. Current
    systems in NYC, Washington DC and Boston have been operated by the
    facility managers for the past several years. This paper presents some
     of the improvements made to the chemical detection system to (a)
    attain a very low detector false positive rate, (b) add new monitoring
     capabilities, and (c) reduce the cost of installation and maintenance
    . Ongoing work includes the addition of a wide range of detection
    devices and extension to biological, radiological, nuclear, and
    explosive detection.},
author = {Policastro, A.J.},
booktitle = {2008 IEEE Conference on Technologies for Homeland Security},
isbn = {1424419778},
keywords = {Automatic control ; Cameras ; Chemicals ; Costs ; Crisis
    management ; Detectors ; Laboratories ; Monitoring ; Protection ;
     Transportation},
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    Management System"},
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@book{alma991972120506254,
abstract = {Evolving threats of chemical, biological, radiological and
    nuclear (CBRN) attack make it imperative to find ways to support
    global efforts against the proliferation of weapons of mass
    destruction (WMD) and terrorism. This book contains papers presented
    at the NATO Advanced Study Institute (ASI) on CBRN security culture,
    held in Yerevan, Armenia, in June 2014. The conference was timely in
    bringing together the various aspects of security culture in the
    different substantive areas from a CBRN perspective, focusing efforts
    to advance CBRN security culture in the everyday work of those who
    must deal with these ever present threats. The primary aim of the
    conference was to promote the concept of a CBRN security culture which
     recognizes synergy across the individual disciplines. As a first step
     in the practical application of this synergistic view, the conference
     introduced assessment methodologies that could be refined and
    customized to enhance the role of the human factor in CBRN security.
    This, we hope, will enable countries to fulfill their international
    obligations and implement United Nations Security Council Resolution
    1540. The book reflects the presentations and deliberations of
    workshop participants and will be of interest to governments,
    international organizations, researchers and practitioners whose work
    involves the human dimension of the security of CBRN materials.},
author = {Thompson, Julia and Gahlaut, Seema},
address = {Amsterdam, [Netherlands},
booktitle = {CBRN security culture in practice},
edition = {1st ed.},
isbn = {1-61499-534-6},
keywords = {Security International ; Terrorism -- Prevention ; Arms
    control},
language = {eng},
organization = {NATO Advanced Study Institute on CBRN Security Culture},
publisher = {IOS Press},
series = {NATO Science for Peace and Security Series. Sub-Series E: Human
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title = {CBRN security culture in practice },
year = {2015},
}
@article{Benghe2026,
author = {Benghe, Cristian and Graure, Vlad and Shreedhar, Tanya and
    Mohan, Nitinder},
address = {New York, NY, USA},
copyright = {Owner/Author},
issn = {2476-1249},
journal = {Proceedings of the ACM on measurement and analysis of
    computing systems},
keywords = {Network performance analysis ;  Network performance modeling
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    ;  Networks},
language = {eng},
number = {2},
pages = {1-30},
publisher = {ACM},
title = {Horizon: Understanding and Predicting Global Starlink
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volume = {10},
year = {2026},
}

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  author       = {{Federal Communications Commission}},
  title        = {Authorization and Order: Space Exploration Holdings,
     LLC (SpaceX Gen2 NGSO Satellite System)},
  number       = {DA 26-36},
  year         = {2026},
  month        = jan,
  day          = {9},
  address      = {Washington, D.C.},
  note         = {Released January 9, 2026. ICFS File Nos. SAT-LOA
     -20200526-00055 et al.}
}
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